Yusuf Berk AKÇAY

Curriculum Vitae

Istanbul, Turkey — yusuf.akcay1@std.bogazici.edu.tr — https://yusufberkakcay52.github.io/

EDUCATION

Mathematics Major with Double Major in Physics 2023 - 2027 (Expected)
Current Year: 3^{rd} year
Institution: Boğaziçi University, Istanbul, Turkey
Current GPA: 3.98/4

RESEARCH INTEREST

I have a broad interest in modern aspects of mathematical physics, especially in differential geometric foundations of various areas of physics, such as classical mechanics, quantization of classical mechanical systems and statistical mechanics. My Github page provides further details on my academic interests and work.

RESEARCH EXPERIENCE

Skeleta in Complex Projective Line March 2026 - Now
Supervisor: Asst. Prof. Umut Varolgüneş

- Studied the holomorphic line bundles of complex projective line with Hermitian structures, their sections and associated Liouville vector fields.
- Working to determine the combinatorial types of the skeleta in terms of the associated section, and relate the intersection properties of two skeleta to the overlap estimates between the corresponding quantum coherent states.

Summer Internship in Florence at ISC-CNR June-August 2026
Supervisor: Prof. Stefano Ruffo

- Studied the ergodic and differential geometric aspects of deriving thermodynamics from classical mechanics, in particular the Rugh formalism and its generalizations, and contributed to the relations between these formalisms.
- Investigated various statistical mechanical models, such as the mean-field ϕ^4 model, its variations, and the one-dimensional XY model by implementing symplectic integrators.

INDEPENDENT STUDIES AND ACADEMIC PROJECTS

PHYS 491: Introduction to Research in Physics I September-December 2024
Supervisor: Asst. Prof. Ilmar Gahramanov

- Studied the Dunkl deformed quantum mechanics with an emphasis on the deformed Pauli oscillator.
- Learned about the Lewis-Riesenfeld Invariant and its uses to obtain exact solutions of the Schrödinger equation.
- Delivered a presentation on the topic to an undergraduate level audience.
- Investigated the possible supersymmetric extensions of the topic.

Course Project, PHYS 177: Computational Introduction to Dynamical Systems July 2025

- Prepared a report on solvable structures and their relation to the Arnold-Liouville theorem.

Reading Program on Geometry of Quantization September 2025 - April 2026
Supervisor: Asst. Prof. Umut Varolgüneş

- Learned about symplectic manifolds, Lagrangian submanifolds, Maslov index and related topics.
- Explored the symplectic geometric formulation of the classical mechanics, with a special emphasis on the Hamilton-Jacobi equation.
- Studied the WKB method to obtain approximate solutions of the Schrödinger equation, geometric interpretation of the WKB solutions and Maslov correction.
- Studied the geometric quantization of general symplectic manifolds.

SCHOOLS AND WORKSHOPS

Istanbul Stringy Meeting
Boğaziçi University, Istanbul, Turkey 2024
COSMOVERSE@ISTANBUL 2025
Istanbul Technical University, Istanbul, Turkey 2025
Grad. and Undergrad. Summer Math School
Nesin Mathematics Village, Izmir, Turkey 2025

- Category Theory
- Conformal Field Theory
- Selected Topics in Modern Physics

37th National Mathematics Symposium
Akdeniz University, Antalya, Turkey 2025
Summer School on the Digital Archive of Exact Solutions in General Relativity
Istanbul Technical University, Istanbul 2025
Istanbul Stringy Meeting
Boğaziçi University, Istanbul, Turkey 2026

TALKS

Workshop on Symmetry and Geometry in Memory of Yavuz Nutku
Hydrodynamic Systems with Two Conserved Quantities 2026

- Integrability and Bi-Hamiltonian structures of d-Boussinesq and Benney-Lax equations

Boğaziçi Math Grad Seminars
WKB Approximation and Geometric Quantization 2026

- Connection between the WKB approximation and the framework of Geometric Quantization

Directed Reading Program 2026
N=4 Super Yang-Mills Theory 2026

- Introduction to Quantum Field Theory, with an emphasis on renormalization, Yang–Mills theory, supersymmetry, and the construction of the N=4 supersymmetric Yang–Mills Lagrangian

HONOURS & AWARDS

- Bronze Medal, TÜBİTAK National Physics Olympiad — 2021
- Ranked Top 0.01, YKS Turkish National University Entrance Exam — 2023
- TÜBİTAK 2205 Undergraduate Scholarship — 2023 - Present

SKILLS

Languages: **Turkish**(native), **English**(fluent)
Computational Tools: **Python, Mathematica, LaTeX**

EXTRACURRICULAR ACTIVITIES

- Club basketball player for 2 years.
- Certified scout for 4 years.
- Organized voluntary lectures for high school students.